Reaction Mechanisms and Catalysis

Elementary steps, catalysts and enzymes

Name _____ Class _____ Date _____

KEY RELATIONSHIPS

Elementary step: rate law follows the coefficients (A + B → … : rate = k[A][B])

Pre-equilibrium: fast A + B ⇌ I gives [I] = K_1[A][B]

Michaelis–Menten: v = Vmax[S] / (Km + [S]) · turnover number kcat = Vmax / [E]total

Enzyme data: Vmax = 150 µmol L^{-1} min^{-1} and Km = 0.50 mmol/L unless stated. Show units in every calculation step.

A. Elementary steps

1. Give the molecularity and the rate law of each elementary step: (a) $O_3 \longrightarrow O_2 + O$ (b) $NO + O_3 \longrightarrow NO_2 + O_2$ (c) $2\,NO_2 \longrightarrow N_2O_4$

2. Hydrogen peroxide decomposes in the presence of iodide ions: step 1 (slow) $H_2O_2 + I^- \longrightarrow H_2O + IO^-$; step 2 (fast) $H_2O_2 + IO^- \longrightarrow H_2O + O_2 + I^-$. (a) Write the overall equation. (b) Identify the catalyst and the intermediate. (c) Write the rate law. (d) Is this homogeneous or heterogeneous catalysis?

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B. Testing mechanisms

3. For $2\,\mathrm{NO} + \mathrm{O_2} \longrightarrow 2\,\mathrm{NO_2}$, rate $= k[\mathrm{NO}]^2[\mathrm{O_2}]$. Mechanism A is one termolecular step. Mechanism B is $2\,\mathrm{NO} \rightleftharpoons \mathrm{N_2O_2}$ (fast), then $\mathrm{N_2O_2} + \mathrm{O_2} \longrightarrow 2\,\mathrm{NO_2}$ (slow). (a) Show that both mechanisms give the experimental rate law. (b) Which is more likely, and why?

4. Ozone decomposes, $2\,\mathrm{O_3} \longrightarrow 3\,\mathrm{O_2}$, by: $\mathrm{O_3} \rightleftharpoons \mathrm{O_2} + \mathrm{O}$ (fast), then $\mathrm{O} + \mathrm{O_3} \longrightarrow 2\,\mathrm{O_2}$ (slow). Derive the rate law, and explain why adding $\mathrm{O_2}$ slows the reaction.

C. Catalysts and enzymes

5. Ethene reacts with hydrogen on a nickel catalyst. Describe the four stages of the catalysed reaction, and explain why the nickel is used as a fine powder.

6. Using the enzyme data, calculate the rate at (a) $[\mathrm{S}] = 0.50$ mmol/L (b) $[\mathrm{S}] = 4.5$ mmol/L. (c) At what [S] is the rate 90 % of $V_{\max}$? (d) At $[\mathrm{S}] = 0.050$ mmol/L, compare the exact rate with the low-[S] approximation $v \approx (V_{\max}/K_{\mathrm{m}})[\mathrm{S}]$.

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D. Challenge

7. Two inhibitors are tested at $[\mathrm{S}] = 1.0$ mmol/L. Inhibitor X is competitive (apparent $K_\mathrm{m} = 1.5$ mmol/L, V_max unchanged). Inhibitor Y is non-competitive (V_max falls to 90 µmol L^{-1} min^{-1}, K_m unchanged). (a) Calculate the rate with no inhibitor, with X and with Y. (b) Explain how measuring the rates at $[\mathrm{S}] = 10$ mmol/L would tell the two inhibitors apart.

8. The enzyme concentration in the experiment is 0.025 µmol/L. (a) Calculate the turnover number in s^{-1}. (b) Explain why the rate of the enzyme reaction first rises and then falls as the temperature is raised from 20 °C to 70 °C.

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